AI Tools and Assistance Document

This document records how AI assistance was used and how generated suggestions were converted into verified work.

1. AI tool used

The primary AI tool was ChatGPT. No technical claim was accepted merely because ChatGPT suggested it.

2. AI assistance log

The log explicitly records: tool used, task requested, generated output, candidate modifications, verification performed, and resulting understanding. Generated output is summarized rather than presented as verbatim prompt history where exact wording was not retained.

Task	Tool	Requested	Generated output	Candidate modifications / verification	Understanding
Synthetic data	ChatGPT	Make the warehouse demand-generating process explicit and reproducible.	A formula/parameterization covering trend, daily and weekly effects, noise, spikes, drops, and a fixed seed.	Executed the generator and checked summary statistics, spike/drop counts, and autocorrelation.	Synthetic data is useful only when its structure and randomness are controlled.
Forecasting dataset	ChatGPT	Express next-hour forecasting as supervised learning with a time-aware split.	Previous 24 hours as input and next hour as target, with chronological train/validation/test splits.	Checked example counts and split boundaries from the actual data; no random temporal mixing.	Temporal forecasting requires time-aware evaluation to avoid leakage.
Mathematical verification	ChatGPT	Structure a central finite-difference gradient check.	Use [L(theta+epsilon)-L(theta-epsilon)]/(2 epsilon), float64, and small epsilon.	Measured relative errors: W_Q 8.221e-11, W_K 5.175e-11, W_V 2.037e-11; stability tests also passed.	Training alone is not proof of mathematical correctness.
Toy learning/scaling	ChatGPT	Design a controlled scaled-vs-unscaled attention comparison.	Same data, model, optimizer, learning rate, epoch count, and seed.	Final MSE: scaled 0.088134; unscaled 0.087712.	Scaling is theoretically motivated, but this experiment did not justify claiming it must lower final loss here.
Distribution shift	ChatGPT	Test a fixed trained model under controlled shifts.	A +40 level shift and doubled noise standard deviation, with no retraining.	Reused the fixed Stage 6 checkpoint and recomputed metrics for each shift.	Generalization can be tested by changing data conditions while keeping the trained model fixed.
Failure investigation	ChatGPT	Inspect the largest individual forecast errors rather than only aggregate metrics.	Top-20 absolute errors, previous demand, target jump, and average final-query attention.	Five largest errors: 211.53, 210.74, 189.82, 184.82, 182.47; associated jumps were about 189–199 units.	Abrupt one-hour changes are a useful diagnostic pattern, but not proof of causality.

3. Corrections and rejected suggestions

Generic Transformer/high-level attention API — Rejected because the assignment required first-principles attention; explicit Q/K/V matrices and softmax calculations were retained.

Claim that attention clearly beat the baselines — Rejected. Persistence had slightly lower MAE (18.5331 vs 18.5462), while attention had lower RMSE (26.5753 vs 30.0543) and slightly lower MAPE (5.30% vs 5.34%).

Claim that scaling always lowers final toy loss — Rejected. The controlled experiment ended at 0.088134 MSE for scaled attention and 0.087712 for unscaled attention.

Attention weights as causal explanations — Rejected. Attention inspection was descriptive; target jumps were treated as diagnostic evidence, not proof of why an individual forecast failed.

Real-world warehouse performance claims — Restricted because the dataset is synthetic; conclusions were limited to the tested synthetic environment and the two tested shifts.

4. Verification discipline

1. Define the requirement before changing the implementation.
2. Implement one focused change at a time.
3. Run targeted tests for the changed behavior.
4. Inspect actual gradients, losses, metrics, generated data, or model outputs.
5. Write interpretations only after the executable evidence is recorded.
6. Run the complete test suite and project-level QA before final submission.

Final QA reported project structure PASS, required generated outputs PASS, result files PASS, and full test suite PASS with 50 tests passed. Reproducibility checks compared repeated same-seed data, loss histories, model parameters, and training normalization statistics.

5. Human responsibility

The final technical responsibility remained with the student: deciding what the assignment required, selecting the small single-head architecture, rejecting out-of-scope abstractions, running the code, checking numerical results, interpreting metrics, investigating failures, and limiting conclusions to the evidence.

6. Final understanding

AI assistance materially improved the speed of requirement decomposition, implementation planning, testing, analysis, and documentation. The key control was keeping AI output subordinate to executable evidence. The project demonstrates that first-principles attention can be implemented and verified directly, while model expressiveness alone does not guarantee a large forecasting advantage over a strong persistence baseline.